

Homework 5

MATH 518 - Fall 2025

Due on Friday, November 21, 11:59pm on Crowdmark

Total of points = 30 points

Please submit each question separately on Crowdmark. Unless stated otherwise, k is an algebraically closed field of characteristic zero.

1. (10 points) Let $X = V(x^3 - y^2) \subset k^2$. Is the map

$$f: k \longrightarrow X, \quad f(t) = (t^2, t^3)$$

a finite map?

Solution: The pullback is the map $f^*: k[X] \longrightarrow k[t]$, where $k[X] = k[x, y]/(x^3 - y^2) = k[\bar{x}, \bar{y}]$, and sends $\bar{x} \mapsto t^2$ and $\bar{y} \mapsto t^3$. It is injective, since the kernel of the morphism $k[x, y] \longrightarrow k[t]$ sending $x \mapsto t^2$ and $y \mapsto t^3$ is exactly $(x^3 - y^2)$. Thus, the image $f^*k[X] = k[t^2, t^3]$ is a subring of $k[t]$. Let us show that $k[t]$ is integral over $k[t^2, t^3]$. The ring $k[t^2, t^3]$ consists of all polynomials in t^{2i+3j} with $i, j \geq 0$. The exponent $n = 2i + 3j$ can be $n = 0$ or any positive integer $n \geq 2$. (If $n \equiv 0, 2 \pmod{3}$ it is clear. If $n = 3j + 1 = 3(j - 1) + 2 \cdot 2$ with $j \geq 1$). Thus, the ring $k[t^2, t^3]$ consists of all polynomial with no term of degree 1, *i.e.* of the form $a_0 + a_1 t^2 h$ for some $h \in k[t]$ and $a_i \in k$. Any polynomial $f \in k[t]$ can be written $f = at + g$ with $g \in k[t^2, t^3]$ and $a \in k$, and we have

$$\begin{aligned} f^2 &= (at + g)^2 = a^2 t^2 + 2atg + g^2 \\ &= a^2 t^2 + 2(f - g)g + g^2 \\ &= 2gf + a^2 t^2 - g^2. \end{aligned}$$

Since $a^2 t^2 - g^2 \in k[t^2, t^3]$, f is integral over $k[t^2, t^3]$.

The vector space of homogeneous polynomials in $k[x_1, \dots, x_{n+1}]$ is of degree $d \geq 1$ is of dimension $N := \binom{n+d}{n}$. It has a basis consisting of monomials $x^I := x_1^{i_1} \cdots x_{n+1}^{i_{n+1}}$, where $I = (i_1, \dots, i_{n+1})$ and $|I| = i_1 + \cdots + i_{n+1} = d$. For $d \geq 1$, we have a morphism

$$\begin{aligned} \nu_d: \mathbb{P}^n(k) &\longrightarrow \mathbb{P}^{N-1}(k) \\ p = [x_1 : \cdots : x_{n+1}] &\longmapsto \nu_d(p) = [x_1^d : x_1^{d-1} x_2 : \cdots : x_{n+1}^d]. \end{aligned}$$

The image consists of all possible monomials x^I of degree d . It is an isomorphism onto its image, called the Veronese embedding. The goal of the next exercise is to show this in the special case $n = 1$ (to simplify the notation, the idea would be the same for $n > 1$).

2. (10 points) When $n = 1$, the Veronese embedding is given by

$$\nu_d([x : y]) = [x^d : x^{d-1}y : \cdots : y^{d-1}x : y^d].$$

Show that ν_d is an isomorphism onto its image, and that the image is the projective variety $V = V_p(I)$ where $I \subset k[z_1, \dots, z_{d+1}]$ is the homogeneous ideal $I := \langle z_i z_j - z_k z_l \mid i + j = k + l \rangle$.

Solution: Let $U_i \subset \mathbb{P}^d(k)$ be the standard open subset $U_i = \{z_i \neq 0\}$. Observe that for a point $[z_1 : \cdots : z_{d+1}] = [x^d : x^{d-1}y : \cdots : y^{d-1}x : y^d] \in \text{im}(\nu_d)$, if z_1 and z_{d+1} are nonzero, then x, y are nonzero and the other z_i 's are also nonzero. So $\text{im}(\nu_d) \subseteq U_1 \cup U_{d+1}$, and $\text{im}(\nu_d) = U'_1 \cup U'_{d+1}$ where $U'_i = \text{im}(\nu_d) \cap U_i$. We define ψ by

$$\psi([z_1 : \cdots : z_{d+1}]) = \begin{cases} [z_1 : z_2] & \text{if } [z_1 : \cdots : z_{d+1}] \in U'_1 \\ [z_d : z_{d+1}] & \text{if } [z_1 : \cdots : z_{d+1}] \in U'_{d+1}. \end{cases} \quad (1)$$

Note that the two agree on $U'_1 \cap U'_{d+1}$ since

$$[z_1 : z_2] = [x^d : x^{d-1}y] = [x : y] = [y^{d-1}x : y^d] = [z_d : z_{d+1}], \quad (2)$$

and it defines a well-defined map $\psi : \text{im}(\nu_d) \rightarrow \mathbb{P}^1(k)$. Moreover, it is a morphism, since the restriction to each subset of the open cover is affine (see homework 4), and it satisfies $\psi \circ \nu_d = \nu_d \circ \psi = \text{id}$. This shows that the Veronese map is an isomorphism onto its image.

For the second part, we have $\text{im}(\nu_d) \subseteq V_p(I)$ since

$$z_i z_j - z_k z_l = (x^{d-i} y^i)(x^{d-j} y^j) - (x^{d-k} y^k)(x^{d-l} y^l) = x^{d-i-j} y^{i+j} - x^{d-k-l} y^{k+l} = 0$$

if $i + j = k + l$. Let us show the other inclusion: $V_p(I) \subseteq \text{im}(\nu_d)$. On $V = V_p(I)$, we have the relation $z_i^2 = z_1 z_{2i-1}$ for $1 \leq i \leq \frac{d}{2}$, and $z_{d+1-i}^2 = z_{d+1} z_{d+1-2i}$ for $1 \leq i \leq \frac{d}{2}$. Thus, if z_i is nonzero for some $2 \leq i \leq d$, then $z_1 \neq 0$ or $z_{d+1} \neq 0$. It follows that

$$V \subseteq U_1 \cup U_{d+1}. \quad (3)$$

Thus, it is enough to show $V \cap U_i \subseteq U'_i = \text{im}(\nu_d) \cap U_i$ for $i = 1$ and $i = d + 1$. Let us show for $i = 1$, the other case is analogous. Suppose that

$$p = [1 : z_2 : \cdots : z_d : z_{d+1}] \in V_p(I) \cap U_{d+1}. \quad (4)$$

Since p is in $V_p(I)$ it satisfies $z_i = z_i z_1 = z_2 z_{i-1}$ for $2 \leq i \leq d + 1$. It follows that $z_3 = z_2^2$, $z_4 = z_2 z_3 = z_2^3$ and so on. This shows that p is of the form

$$p = [1 : z_2 : z_2^2 \cdots : z_2^{d-1} : z_2^d] \in U'_1. \quad (5)$$

3. (10 points) For a homogeneous polynomial $f \in k[x_1, \dots, x_{n+1}]$ of degree $d \geq 1$, let $U_f \subseteq \mathbb{P}^n(k)$ be the open subset

$$U_f := \{p \in \mathbb{P}^n(k) \mid f(p) \neq 0\} = \mathbb{P}^n(k) \setminus V_p(f).$$

In the last lecture, we have seen that U_f is an affine variety when $f = a_1x_1 + \dots + a_{n+1}x_{n+1}$ is a linear polynomial. Use it to show that U_f is affine for any f .

Hint: use the Veronese embedding.

Solution: Suppose that $f = \sum_I a_I x^I \in k[x_1, \dots, x_{n+1}]$ is a polynomial of degree $d \geq 1$. Let $L \in k[z_1, \dots, z_N]$ be the linear polynomial

$$L = \sum_I a_I z_I$$

in the variables z_I , indexed by the sets I . Observe that if $p \in V_p(f)$ is a zero of f , then $\nu_d(p)$ is a zero of $V_p(L)$. In fact, we have

$$\nu_d(\mathbb{P}^n(k)) \cap V_p(L) = \nu_d(V_p(f)). \quad (6)$$

It follows that

$$\begin{aligned} \nu_d(U_f) &= \nu_d(\mathbb{P}^n(k) \setminus V_p(f)) \\ &= \nu_d(\mathbb{P}^n(k)) \setminus \nu_d(V_p(f)) \\ &= \nu_d(\mathbb{P}^n(k)) \setminus \nu_d(\mathbb{P}^n(k)) \cap V_p(L) \\ &= \nu_d(\mathbb{P}^n(k)) \cap U_L, \end{aligned}$$

where $U_L \subseteq \mathbb{P}^{N-1}(k)$ is the open set given by the L . It is affine, thus isomorphic to a closed subspace of some k^r . Since $\nu_d(\mathbb{P}^n(k))$ is closed (image of projective variety by regular map is closed), it follows that $U_f \simeq \nu_d(U_f) = \nu_d(\mathbb{P}^n(k)) \cap U_L$ is closed in U_L , thus also a closed affine set.